Introduction To Flight Mechanics

Tutorial 7



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Exercise 7: Key Facts

An aircraft with a turbojet propulsion shall fly a distance of s = 2000 km with a velocity of V = 65 m/s with the least amount of fuel. Throughout the flight, the lift coefficient is constant. Following data is given:

Description	Parameter	Value
Aircraft mass (including fuel) at the beginning of the flight	m	$2000\ \mathrm{kg}$
Drag factor	k	$0{,}018$
Wing reference area	S_{ref}	$18\ \mathrm{m}^2$
Drag coefficient	C_{D0}	$0{,}01$
Specific fuel consumption with respect to the engine performance parameter	b_L	$7{,}6 \cdot 10^{-6}\,\frac{kg}{N}$

Key Facts ISA

MSL density ISA standard atmosphere: $\rho_{s,MSL} = 1.225 \frac{kg}{m^3}$

MSL pressure ISA standard atmosphere: $p_{s,MSL} = 1.01325e5\ Pa$

MSL temperature ISA standard atmosphere: $T_{s,MSL} = 288.15\ K$

Gravitational acceleration: $g_s = 9.80665 \frac{m}{s^2}$

Speed of sound MSL: $a_{s,MSL} = 340 \frac{m}{s}$

Specific gas constant: $R = 287.05 \frac{J}{kg\ K}$

Polytropic exponent: $n = 1.235$

$\kappa = 1{,}4$

Different Types of Cruise

$$dt = -\frac{V^{n_v}}{g\ b_L}\frac{C_L}{C_D}\frac{dm}{m}$$

$$ds = -\frac{V^{n_v+1}}{g\ b_L}\frac{C_L}{C_D}\frac{dm}{m}$$

Condition for lift:

$$\frac{1}{2}\rho\, V^2\, S\, C_L = m\, g$$

→ Three equations for cruise

⟹ 3 independent variables C_L, V, ρ

Integration of the first two equations yields the flight time and range. In order to simplify the integration, two of the three variables V, ρ und C_L are set to a constant value. The third quantity results from the lift equation.

The following combinations are possible:

Type	V	C_L	ρ	Variable
I	const.	const.	-----	$\rho \sim m$
II	const.	-----	const.	$C_L \sim m$

Annotation: The third possibility of changing velocity is not part of this class.

Exercise 7.1

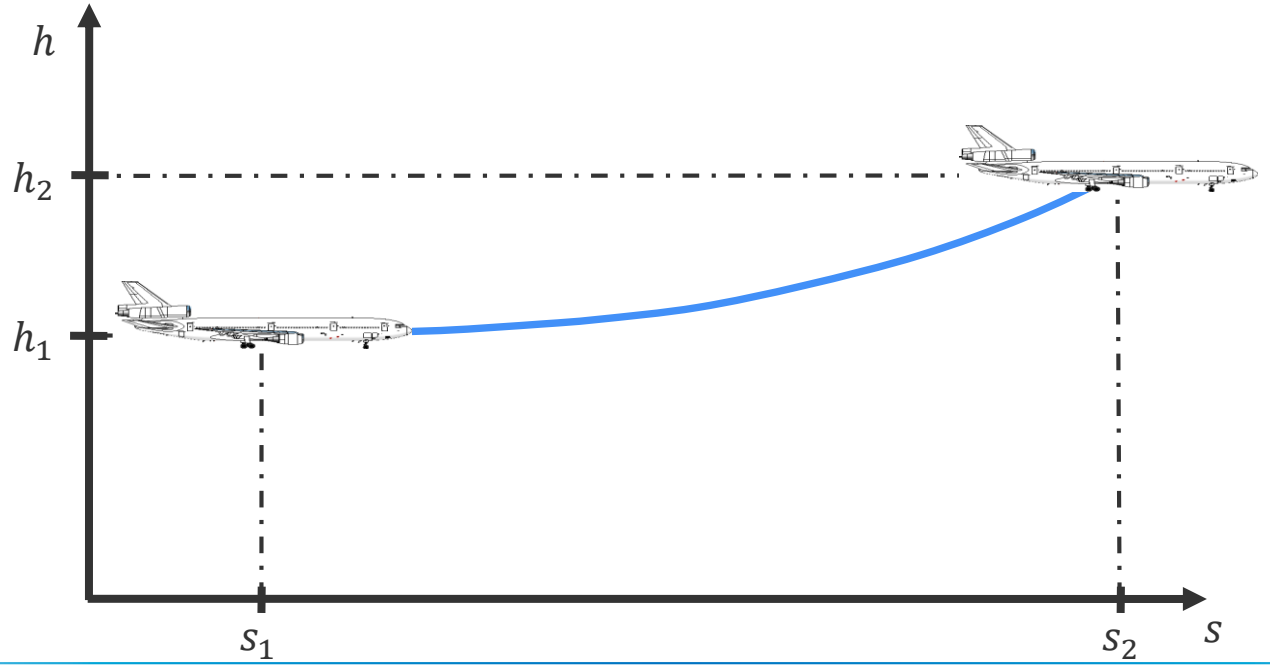
What is the initial and the final height of the aircraft? How long does the flight take? Assume a symmetric quadratic drag polar.

Which type of cruise do we investigate?

*An aircraft with a turbojet propulsion shall fly a distance of s = 2000 km with a velocity of **V = 65 m/s** with the **least amount of fuel**. Throughout the flight, the lift coefficient is constant.*

Type I – Velocity *V* and coefficient of lift C_L as constants

- Flight time: $t = \int_{m_1}^{m_2} -\frac{V^{n_v}}{g\, b_L}\frac{C_L}{C_D}\frac{dm}{m} = \frac{V^{n_V}}{g\, b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2}$
- Range: $s = \int_{m_1}^{m_2} -\frac{V^{n_v+1}}{g\, b_L}\frac{C_L}{C_D}\frac{dm}{m} = \frac{V^{n_V+1}}{g\, b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2}$
- Direct integration of the equations, as C_L and V are constant and can be put in front of the integration



What is the initial and the final height of the aircraft? How long does the flight take? Assume a symmetric quadratic drag polar.

$$s = \int \frac{V^{n_v+1}}{g\ b_L} \frac{C_L}{C_D} \frac{dm}{m} = \frac{V^{n_V+1}}{g\ b_L} \frac{C_L}{C_D} \ln \frac{m_1}{m_2}$$

Which lift factor C_L do we choose for the cruise flight?

As demonstrated on the next slide, this depends on:

a) The propulsion system, and
b) The fact whether a cruise velocity V is given

Which lift factor C_L do we choose for the cruise flight?

$$s = \int \frac{V^{n_v+1}}{g\, b_L} \frac{C_L}{C_D} \frac{dm}{m} = \frac{V^{n_V+1}}{g\, b_L} \frac{C_L}{C_D} \ln \frac{m_1}{m_2}$$

Lift condition

$$\frac{1}{2} \rho\, V^2\, S_{ref} C_L = m\, g$$

piston engine with $n_V = -1$: $s \propto \frac{C_L}{C_D}$ max s ⇨ $C_L^* = \sqrt{\frac{C_{D0}}{k}} \rightarrow V^*$

turbojet engine with $n_V = 0$: $s \propto V \frac{C_L}{C_D}$ max s ⇦ *Relevant for this task*

a) If V given: $V = \text{const}$ and not dependent on C_L

b) If V not given:

$$s \propto V \frac{C_L}{C_D} = \sqrt{\frac{2mg}{\rho S_{ref} C_L}} \frac{C_L}{C_D} = \sqrt{\frac{2mgC_L}{\rho S_{ref}}} \frac{1}{C_D} \propto \underbrace{\frac{\sqrt{C_L}}{(C_{D,0} + kC_L^2)}}_{f(C_L)}$$

max s ⇦ $\frac{df(C_L)}{C_L} = \cdots = \frac{3kC_L^2 - C_{D,0}}{\sqrt{C_L}(C_{D,0} + kC_L^2)}$

$$C_L = \sqrt{\frac{C_{D0}}{3k}} = \frac{C_L^*}{\sqrt{3}}$$

What is the **initial** and the final **height of the aircraft**? How long does the flight take? Assume a symmetric quadratic drag polar.

We know V^*. Can we compute the corresponding (initial) altitude?

$$\underbrace{\frac{1}{2}\rho V^2 S_{ref} C_L = m\,g}_{\text{Horizontal flight}} \rightarrow \quad \rho = \frac{2 \cdot m\,g}{V^2 S_{ref}\,C_L}$$

$$\Rightarrow \quad \boxed{\rho_1 = \frac{2 \cdot m_1\,g}{V^{*2} S_{ref}\,C_L^*} = 0{,}69\,\frac{kg}{m^3}}$$

What is the **initial** and the final **height of the aircraft**? How long does the flight take? Assume a symmetric quadratic drag polar.

We know V^*. Can we compute the corresponding (initial) altitude?

$$\rho_1 = \frac{2 \cdot m_1 \, g}{V^{*2} S_{ref} \, C_L^*} = 0{,}69 \frac{kg}{m^3}$$

We can now compute the corresponding altitude using the standard atmosphere equations:

$$H_{G1} = \frac{\left(\left(\frac{\rho}{\rho_{S,MSL}} \right)^{n-1} - 1 \right)}{\gamma_{TR}} \cdot T_{S,MSL} = 5568{,}7 \text{ m}$$

$$h_{geod1} = \frac{r_{earth} \cdot H_{G1}}{r_{earth} - H_{G1}} = 5573{,}6 \, m$$

What is the initial and the **final height of the aircraft**? How long does the flight take? Assume a symmetric quadratic drag polar.

$$s = \int \frac{V^{n_v+1}}{g\ b_L}\frac{C_L}{C_D}\frac{dm}{m} = \frac{V^{n_V+1}}{g\ b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2}$$

$$\Rightarrow \quad m_2 = \frac{m_1}{\exp\left(\frac{s \cdot g \cdot b_L \cdot C_D^*}{V^{*n_V+1} \cdot C_L^*}\right)} = 1880{,}6\ kg$$

We can now compute the corresponding density at time *2*:

$$\rho_2 = \frac{2 \cdot m_2\ g}{V^{*2} S_{ref}\ C_L^*} = 0{,}65\,\frac{kg}{m^3}$$

What is the initial and the **final height of the aircraft**? How long does the flight take? Assume a symmetric quadratic drag polar.

We know V^*. Can we compute the corresponding (final) altitude?

$$\rho_2 = \frac{2 \cdot m_2\, g}{V^{*2} S_{ref}\, C_L^*} = 0{,}65\,\frac{kg}{m^3}$$

We can now compute the corresponding altitude using the standard atmosphere equations:

$$H_{G2} = \frac{\left(\left(\frac{\rho_2}{\rho_{S,MSL}}\right)^{n-1} - 1\right)}{\gamma_{TR}} \cdot T_{S,MSL} = 6125{,}4\ \mathrm{m}$$

$$h_{geod2} = \frac{r_{earth} \cdot H_G}{r_{earth} - H_G} = 6131{,}3m$$

What is the initial and the final height of the aircraft? **How long does the flight take**? Assume a symmetric quadratic drag polar.

$$t_{flight} = \frac{V^{*n_V}}{g\ b_L}\frac{C_L}{C_D}\ln\frac{m_1}{m_2} = 3{,}07 \cdot 10^4\ seconds = 8{,}54\ hours$$

Exercise 7.1 – numbers A320

Description	Parameter	Value
Aircraft mass (including fuel) at the beginning of flight	m	65000 kg
Drag factor	k	0,036
Wing reference area	S_{ref}	122,6 m^2
Drag coefficent	C_{D0}	0,024
Velocity	V	200 m/s
Specific fuel consumption with respect to propulsion performance parameter	b_L	$1{,}7 \cdot 10^{-5} \frac{kg}{N}$
Distance	s	2000km

$$H_{G1} = 12057 \text{ m} \qquad H_{G2} = 12796 \text{ m} \qquad \Delta H_G = 738 \text{ m}$$

$$m_2 = 58932\ kg \qquad \Delta m = 6068\ kg$$

$$t_{flight} = 1{,}00 \cdot 10^4\ seconds = 2{,}78\ h$$

Errors due to $n_V = 0$

Exercise 7.2: Key Facts

A Turbojet-aircraft shows following characteristics

Description	Parameter	Value
Maximum engine performance parameter	L_{Pmax}	5000 N
Lift coefficient	C_{L0}	0,01
Lift derivative	$C_{L\alpha}$	6,2
Aircraft mass	m	2000 kg
Drag factor	k	0,02
Wing reference area	S_{ref}	18 m^2
Drag coefficient	C_{D0}	0,02
Maximum lift coefficient	C_{Lmax}	1,2

For a true airspeed of $\boldsymbol{V_{TAS} = 65\frac{m}{s}}$ the pilot would like to start a steady state climb (straight and without sideslip) with $\gamma_K = 3°$. He puts the thrust lever to $\delta_T = 0.5$. What is the resulting angle of attack? Show the solution via the rigid body equations of motion.

The RBE before inserting the forces:

$$\dot{V}_K^G = \frac{\left(X_T^G\right)_{\bar{K}}}{m}$$

$$\dot{\alpha}_K = \frac{\left(Z_T^G\right)_{\bar{K}}}{m \cdot V_K^G \cdot \cos\beta_K} + q_K - \tan\beta_K \cdot (p_K \cdot \cos\alpha_K + r_K \cdot \sin\alpha_K)$$

$$\dot{\beta}_K = \frac{\left(Y_T^G\right)_{\bar{K}}}{m \cdot V_K^G} + (p_K \cdot \sin\alpha_K - r_K \cdot \cos\alpha_K)$$

Inserting the forces:

$$\dot{V}_K^G = \frac{T - D}{m} - g \sin\gamma$$

$$\dot{\alpha}_K = \frac{-L}{m \cdot V_K^G \cdot \cos\beta_K} + \frac{g}{V} \cdot \frac{\cos\mu \cos\gamma}{\cos\beta_K} + [q_K - \tan\beta_K \cdot (p_K \cdot \cos\alpha_K + r_K \cdot \sin\alpha_K)]$$

$$\dot{\beta}_K = \frac{Q}{m \cdot V_K^G} + \frac{g}{V} \cos\gamma \sin\mu + (p_K \cdot \sin\alpha_K - r_K \cdot \cos\alpha_K)$$

Now lets simplify:

$$\dot{V}_K^G = \frac{T - D}{m} - g \sin\gamma \qquad (=0)$$

$$\dot{\alpha}_K = \frac{-L}{m \cdot V_K^G \cdot \cos\beta_K} + \frac{g}{V} \cdot \frac{\cos\mu \cos\gamma}{\cos\beta_K} + [q_K - \tan\beta_K \cdot (p_K \cdot \cos\alpha_K + r_K \cdot \sin\alpha_K)]$$

$(=0)$ $(=1)$ $(=1)$ $(=1)$ $(=0)$ $(=0)$ $(=0)$ $(=0)$

$$\dot{\beta}_K = \frac{C}{m \cdot V_K^G} + \frac{g}{V}\cos\gamma \sin\mu + (p_K \cdot \sin\alpha_K - r_K \cdot \cos\alpha_K)$$

$(=0)$ $(=0)$ $(=0)$ $(=0)$ $(=0)$

"steady state climb (straight and without sideslip)"

For a true airspeed of $\boldsymbol{V_{TAS} = 65\frac{m}{s}}$ the pilot would like to start a steady state climb (straight and without sideslip) with $\gamma_K = 3°$. He puts the thrust lever to $\delta_T = 0.5$. What is the resulting angle of attack? Show the solution via the rigid body equations of motion.

$$0 = \frac{T - D}{m} - g \sin \gamma_K$$

$$\boxed{0 = \frac{-L}{m \cdot V_K^G} + \frac{g}{V_K^G} \cdot \cos \gamma_K}$$

$$\dot{\beta}_K = 0$$

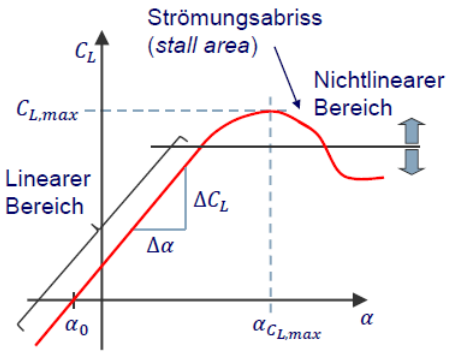
For a true airspeed of $\boldsymbol{V_{TAS} = 65\frac{m}{s}}$ the pilot would like to start a steady state climb (straight and without sideslip) with $\gamma_K = 3°$. He puts the thrust lever to $\delta_T = 0.5$. What is the resulting angle of attack? Show the solution via the rigid body equations of motion.

Let‘s find the α:

$$0 = \frac{-L}{m \cdot V_K^G} + \frac{g}{V_K^G} \cdot \cos\gamma_K$$

$$L = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot C_L$$

Recall the definition of the lift coefficient C_L:



$$C_L \approx C_{L0} + \underbrace{\frac{\Delta C_L}{\Delta\alpha}}\ \alpha$$

$$C_L \approx C_{L0} + C_{L\alpha} \cdot \alpha$$

After inserting the definition of the lift coefficient, we end up with:

$$0 = \frac{-\rho \cdot V^2 \cdot S_{ref} \cdot \overbrace{(C_{L0} + C_{L\alpha} \cdot \alpha)}^{C_L}}{2 \cdot m \cdot V} + \frac{g}{V} \cdot \cos\gamma$$

⇨ two unknowns: α and ρ

We therefore need a second equation ($\dot{V}_K^G$ equation):

$$0 = \frac{T - \overbrace{\frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot (C_{D0} + k \cdot (C_{L0} + C_{L\alpha} \cdot \alpha)^2)}^{D}}{m} - g \sin\gamma_K$$

We can compute T: $T = L_P \cdot V^{n_V}$ mit $n_V = 0 \rightarrow T = L_P$

$$T = L_P = L_{Pmax}(\rho) * \delta_T = 5000 \cdot 0{,}5 = 2500\ N$$

After inserting the definition of the lift coefficient, we end up with:

$$0 = \frac{-\rho \cdot V^2 \cdot S_{ref} \cdot \overbrace{(C_{L0} + C_{L\alpha} \cdot \alpha)}^{C_L}}{2 \cdot m \cdot V} + \frac{g}{V} \cdot \cos\gamma$$

⇨ two unknowns: α and ρ

We therefore need a second equation ($\dot{V}_K^G$ equation):

$$0 = \frac{T - \overbrace{\frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot (C_{D0} + k \cdot (C_{L0} + C_{L\alpha} \cdot \alpha)^2)}^{D}}{m} - g \sin\gamma_K$$

And then D: $T - D = mg \cdot \sin\gamma \rightarrow D = T - mg \cdot \sin\gamma$

$$D = [2500 - 2000 * 9.80665 \cdot \sin 3°]N = 1473{,}2\ N$$

Exercise 7.2

$$0 = \frac{-\rho \cdot V^2 \cdot S_{ref} \cdot (C_{L0} + C_{L\alpha} \cdot \alpha)}{2 \cdot m \cdot V} + \frac{g}{V} \cdot \cos\gamma \qquad D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot (C_{D0} + k \cdot (C_{L0} + C_{L\alpha} \cdot \alpha)^2)$$

Combining equations leads to :

$$\underbrace{k \cdot mg \ cos\gamma}_{a} \ C_L^2 \underbrace{- D}_{b} C_L + \underbrace{mg \ cos\gamma \ C_{D0}}_{c} = 0$$

⇨ Solve quadratic equation for C_L. Note, we know D!

~~$C_{L,1} = 3{,}4713$~~ Not physical when comparing to C_{Lmax}

$C_{L,2} = 0{,}2881$

$$C_{L,2} = 0{,}2881$$

$$(C_{L0} + C_{L\alpha} \cdot \alpha) = C_{L,2} \rightarrow$$

$$\boxed{\alpha = \frac{C_{L,2} - C_{L0}}{C_{L\alpha}} = 0.0449rad = 2{,}5698°}$$